AMARNATH S. PATEL

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Scientific software — numerical methods, GPU acceleration, and verification you can actually check.

EDUCATION

University of Central Florida **Undergraduate Student 4.00 GPA**
B.S. Physics (Optics & Lasers) and Mathematics, Computer Science minor August 2025 – Expected May 2029

- **Mathematics:** Linear Algebra (proof-based, in progress), Complex Analysis, Partial Differential Equations (in progress), Applied Linear Algebra, Ordinary Differential Equations, Mathematical Methods for Physics
- **Computing:** Data Structures & Algorithms, Computer Architecture, Computer Logic Design, Object-Oriented Programming, Discrete Structures, Deep Learning

Florida Atlantic University **University coursework via FAU High School, ages 14–18 3.66 GPA**
111 credit hours completed between ages 14 and 18, alongside high school August 2021 – May 2025

SCIENTIFIC SOFTWARE

CELERIS — electromagnetic solver and metasurface design pipeline. C++23, CUDA, Python, CMake 2026

- Wrote a full rigorous coupled-wave analysis solver from scratch rather than wrapping an existing kernel: per-layer Fourier-basis eigenproblems, dense complex linear algebra, Li and Liu–Fan inverse-rule factorization, the 2D-vectorial $P \cdot Q$ formulation for biperiodic structures, and a Redheffer scattering-matrix recursion chosen over the transfer-matrix form for numerical stability across many layers.
- Treated verification as a first-class feature: every layer is checked against closed-form physics, an independent method, or energy conservation — 1e-6 against the transfer-matrix method, 1e-15 on multilayer recursion, 5e-12 on a degenerate-case reduction, Σ (diffraction efficiency) = 1.000000 — and the assembled solver is cross-validated against the grcwa and Stanford S⁴ codes to ≈ 1e-7, reproducing canonical published results (Khorasaninejad et al., *Science* 2016; Chen et al., *Nat. Nanotechnol.* 2018).
- Implemented Rayleigh–Sommerfeld far-field propagation as a CUDA kernel and benchmarked it honestly — against the project's own optimized 16-core CPU path, not against a naive baseline: 4.8–5.8× at 92k elements, agreeing to machine precision, with automatic CPU fallback. Diagnosed the kernel as memory-bound rather than compute-bound from the way the speedup declined as the grid grew.
- Built the science on top of the kernel: material dispersion (Sellmeier and tabulated n, k), parameter sweeps, gradient-based (Adam) inverse design, Monte-Carlo tolerance analysis, a wavefront/MTF/Strehl analysis battery, and GDSII export. Python bindings and a native Qt desktop GUI over the C++ core. *Sole-author manuscript in preparation (Computer Physics Communications).*

Stardust — archival recovery of near-Earth asteroids. PyTorch, CUDA, Python October 2026

- Built the orbit and detection pipeline: a batched GPU N-body integrator in PyTorch (Sun, planets, Moon, 16 massive asteroids from JPL DE440, solar relativity) samples the orbit uncertainty of a 4-day-old asteroid, 100,000 clones × 1 year in 23 s on a laptop RTX 4070. Validated against JPL on 2024 YR4: within 4 km of REBOUND+ASSIST after a year, 2032 Earth miss 279,443 km versus JPL's 278,333 km.
- Projected the clone cloud onto years of archival ZTF images and searched with a U-Net detector and a matched streak filter, keeping only tracks that survive an orbit refit. Blind backtest on 72 once-risky asteroids: 167 verified, never-connected sightings of 22, tightening orbits up to 52× (median 7.1×).

Undergraduate Researcher — UCF Astrophotonics Lab, CREOL August 2025 – Present
Supervisor: Dr. Stephen Eikenberry

- Built the lab's data-acquisition system: a Python stack that drives four instruments as one coordinated experiment across GPIB/VISA, RS-232 and GigE Vision, with in-loop optimization of a live objective and automatic rejection of saturated frames. Reduced a full measurement campaign from days of manual operation to a single unattended run.
- Implemented the reduction pipeline that turns raw interferograms into physics: FFT sideband isolation and demodulation, Butterworth filtering, joint nonlinear optimization over three coupled parameters, and decomposition onto an analytic mode basis (NumPy/SciPy). Characterized at ≈ 98% reconstruction fidelity against simulated ground truth.
- Own the node software for PolyOculus, a networked array of 8 telescopes: brought up mainline Linux on the array's Jetson Nanos to replace NVIDIA's end-of-life image (custom kernel, no initramfs, Ansible-managed fleet), and wrote centroid-based autoguiding to INDI mount guide pulses that cuts simulated tracking error from 3.01″ to 0.29″ RMS, plus a PI-loop piezo steering mirror holding a laser spot to 0.2 px.

Undergraduate Research Assistant — UCF Physics Department March 2026 – Present
Physics Education Research — Dr. Zhongzhou Chen (NSF Award 2421299; Gates Foundation INV-076932)

- Built ESTELA (Rust, Tauri 2), a research tool generating multi-version isomorphic exams from a structured bank of 615 problems across 29 banks, 13 topic areas and 11 question types, with export to four formats. In production use by external adopters outside the university; manuscript in preparation.
- Built an agentic generation–evaluation pipeline over the Azure OpenAI batch API that drafts and validates candidate problems against instructor criteria.

Other software 2023 – Present

- Systems and tooling in C, Rust and Go: hush, an interactive shell in C; limebar, a Wayland status bar in C; constrict, a build system in Go written to replace make; coat, a universal theme applier in Rust.

TECHNICAL SKILLS

Numerical methods: dense and complex linear algebra, eigenvalue problems, FFT and signal processing, gradient-based optimization, Monte-Carlo methods, phase retrieval and modal decomposition, error analysis and convergence testing

High performance: CUDA kernel development and profiling, multicore CPU optimization, memory-bound versus compute-bound analysis, HPC cluster work, embedded Linux on NVIDIA Jetson

Languages and tools: C++ (C++23), CUDA, C, Python (NumPy, SciPy), Rust, Go; Linux, Git, CMake, Docker, Qt/PySide6, LaTeX/Typst; GPIB/VISA, RS-232, GigE Vision, FITS, GDSII

HONORS

Florida Bright Futures — Florida Academic Scholars (highest tier; 100% tuition) 2025
Lockheed Martin Award, "Highest Level of Engineering Excellence," AEV Competition 2024